

TixelERP Development Progress Report

Full platform review against the shared tracking sheet – 102 tracked features across 13 modules

CLIENT COPY

PREPARED BY

Biswajit,
Trackepay

PREPARED FOR

Tixel

DATE

14 July 2026

SCOPE

All 13 modules

00 Executive summary

This report replaces status claims with verified ones. Every row below was checked directly against the running application — its data structures, business logic, and the actual page behaviour — not inferred from module names or the tracking sheet's own remarks. Where the sheet said "done" and the application disagreed, the application wins, and the discrepancy is called out.

48

DONE — 47%

32

PARTIAL — 31%

22

NOT STARTED — 22%

■ Done — built, wired to real data, no missing dependency

■ Partial — real code exists but a meaningful piece is stubbed, simulated, or unwired

■ Not started — no corresponding model, action, or route found

What's solid: the core system-of-record — Finance ledger/bills/payroll, HRM (leaves, performance, scheduling, fleet), full Office suite (Word/Excel/PowerPoint-alikes, real IMAP/SMTP webmail, WebRTC calls), Organization (departments, branches, dynamic RBAC down to the designation level), and POS (billing, captain workflow, table manager, loyalty tokens, QR ordering) are genuinely built against the database, not mocked. The dynamic Designation/RBAC engine matches the sheet's own note in the *RBACDesignation* tab almost exactly.

What needs attention first: three things repeat across nearly every "Partial" row — (1) features that generate a document by calling the browser's print dialog instead of a server-side PDF, (2) integrations where the OAuth/API client code is written but never actually invoked (Google Calendar sync, campaign sending), and (3) files (contracts, signatures, fuel-log photos) stored as base64 text directly in Postgres with no size limit, which will not hold up at scale. None of these are large rewrites, but they should be sequenced before new feature work.

What's genuinely not started: the Report Generator / Civil Module (Geotechnical, Survey, Design report calculators) is a template shell with no domain logic; statutory/GST filing and government portal integration; SMS marketing; and there is no mobile app anywhere in the platform today — which blocks geofenced attendance, live HR/asset map tracking, and biometric login as specified.

1.0 Login / Signup

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Authentication (NextAuth, credentials + OAuth2)	DONE	NextAuth v5, bcrypt password hashing, Google & Microsoft Entra OAuth wired, account lockout after 5 failed attempts, rate limiting, audit log.	
Two-factor authentication (TOTP)	DONE	Not in original spec, but real: encrypted TOTP secret + hashed recovery codes.	
Forgot password flow	PARTIAL	Reset-completion step works end to end. The "request a link" screen has an unfinished handler (fake timeout, never calls the API) and the API itself only logs the reset link server-side instead of emailing it.	
Session time tied to shift schedule	NOT STARTED	No link between an employee's assigned shift and their session expiry time.	
Single-device login enforcement	NOT STARTED	No device fingerprinting or forced-logout-from-other-device flow found.	
Biometric login (mobile)	NOT STARTED	Moot — no mobile app exists yet.	MOBILE APP

2.0 Overview / Dashboard

FEATURE	STATUS	EVIDENCE	DEPENDENCY
5-tab structure (Project / Finance / Attendance / Resources / Map)	DONE	All five tabs render against real, live queries — no placeholder tabs.	
Project / Finance / Attendance / Resources card data	DONE	Live query data, not placeholders.	
Cards open their linked page on click	NOT STARTED	Four dashboard tab components contain zero navigation links — cards display data but don't drill down.	
Map — live HR / asset movement tracking	NOT STARTED	The map itself is a hand-drawn SVG placeholder over hardcoded sample coordinates; the UI text literally reads "OSM integration pending." No live location ingestion endpoint exists.	MOBILE APP

3.0 Finance

Day-to-day bookkeeping (ledger, bills, payroll, payments) is solid. The advanced statutory/compliance layer the sheet asks for — GST filing, TDS, fixed-asset depreciation, treasury, multi-company consolidation — has not been started.

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Books — bank accounts + statement reconciliation	DONE	Real CSV bank-statement upload, parse, and reconcile flow.	
Journal	DONE	Post/void journal entries against General Ledger accounts.	
Expenses (with project linkage)	DONE	Approval workflow, optional project assignment.	
Payroll / payslip generation	PARTIAL	Real statutory computation (basic/HRA/DA splits) exists, but the client feedback log has an unresolved report that "payslip generation is not functioning" — needs re-verification with the reporter before calling this closed.	

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Bills / Settle (vendor bills)	DONE	Create, approve, and pay vendor bills end to end.	
Credit Notes	DONE	Full CRUD.	
Payments (gateway)	PARTIAL	Razorpay + Stripe SDKs genuinely wired with signature verification on create/verify/webhook.	THIRD-PARTY API
Financial reports	PARTIAL	Trial balance + P&L/balance-sheet generation works; sheet itself flags this "TBD" pending Tixel-side review.	
Financial documents	DONE	Upload + keyword search.	
General Ledger / Chart of Accounts	DONE	Basic GL functioning under Books.	
Accounts Receivable — GST invoicing, e-invoice/e-way bill	PARTIAL	CGST/SGST/IGST fields present on invoices; IRN e-invoicing and e-way bill generation not implemented.	GOVT PORTAL
Accounts Payable — vendor TDS engine (Form 16A/26Q)	NOT STARTED	No TDS rate computation or form generation found.	
GST & Indirect Tax Engine (GSTR-1/3B/9)	NOT STARTED	No GSTN filing client anywhere.	GOVT PORTAL
Bank & Cash Management (auto import)	PARTIAL	Manual CSV upload works; no live bank-feed API.	
Fixed Assets & Depreciation	NOT STARTED	Asset records track cost and current value only — no depreciation method or schedule.	
Budgeting & FP&A	NOT STARTED	No budget model.	
Treasury / Cash & Liquidity	NOT STARTED	No forecasting logic.	
Multi-Company Consolidation	NOT STARTED	Multi-tenant data isolation exists; multi-entity consolidation/eliminations do not.	

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Regulatory / Govt Portal Integration	NOT STARTED	No GSTN, ICEGATE, MCA21, or EPFO/ESIC client code.	GOVT PORTAL
Credit Management & Collection (bureau feeds)	NOT STARTED	No credit-bureau integration.	THIRD-PARTY API

4.0 Sales & POS

FEATURE	STATUS	EVIDENCE	DEPENDENCY
CRM / Tenders & Leads	DONE	Extensive CRUD across Leads, Tenders, Bids, EMD, BOQ.	
Client / Vendor database	DONE	Contact records with geocoding.	
Lead → Project pipeline conversion	PARTIAL	Kanban pipeline stages are real; conversion lands on a Contact/Deal, not directly on a Project record.	
Simulation ("what-if" projections)	NOT STARTED	No scenario/simulation logic found anywhere in the platform.	
Quotations	PARTIAL	CRUD + GST fields are real. "PDF" is the browser print dialog, not a server-rendered file; the digital-sign step is a labeled "signature pad simulation" overlay on the print view, not a legal e-signature.	
Invoices / Running Account Bills	PARTIAL	Standard invoicing works; no progressive/RA-bill-specific billing logic for construction work-done percentages.	
Subscriptions (recurring invoices)	PARTIAL	Recurrence fields and CRUD exist; no reminder cron job runs against them.	
Sales Visit / Route	PARTIAL	Genuine nearest-neighbour route optimization on an in-app Leaflet map; does not hand off to Google/Apple Maps as specified.	
POS — billing	DONE	Order + item CRUD, status flow.	
POS — order integration (Zomato / Swiggy)	NOT STARTED	No references anywhere in source.	THIRD-PARTY API

FEATURE	STATUS	EVIDENCE	DEPENDENCY
POS — captain features	DONE	Explicitly implemented and labeled as such in code.	
POS — table manager	DONE	Reservation CRUD + status flow.	
POS — token / points management	DONE	Queue tokens and loyalty points both functional.	
POS — QR generator	DONE	Menu, UPI-payment, and feedback QR presets.	
POS — kiosk software	NOT STARTED	No matches anywhere.	HARDWARE
POS — waiter calling system	NOT STARTED	No matches anywhere.	

5.0 Marketing

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Email marketing	PARTIAL	Campaign UI and stats are fully DB-backed, but the send function is explicitly labeled "simulated sending" — it writes random stats instead of dispatching real mail (the working SMTP client is only wired to the separate webmail module).	
SMS marketing	NOT STARTED	Channel selectable in UI but falls through to the same simulated path; no SMS gateway SDK anywhere.	THIRD-PARTY API
WhatsApp marketing	DONE	Real Meta WhatsApp Cloud API client, genuinely called per-contact for campaigns. Inbound webhook handshake works but doesn't persist incoming messages yet.	THIRD-PARTY API
Social marketing	PARTIAL	Real Facebook/LinkedIn/Twitter posting clients exist (one-way publish only); no visitor-analytics tracking.	THIRD-PARTY API
Events (publish + sell tickets)	PARTIAL	Event/attendee CRUD with a ticket-price field, but no payment gateway call in the registration flow — nothing is actually charged.	THIRD-PARTY API
Survey	PARTIAL	Full builder + analytics exist internally, but there's no public-facing route for a respondent to actually open and fill in a survey link.	

6.0 Site Store / Supply Chain

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Inventory & Stock (merged)	DONE	Products, stock movement, valuation, low-stock alerts, spreadsheet bulk-upload.	
Warehouse	DONE	Manager/contact/location fields all wired.	
Assets / Maintenance	PARTIAL	Preventive/corrective maintenance requests work; no insurance field, no spares-used tracking.	
Manufacturing (orders & BOM)	DONE	Real order + bill-of-materials CRUD.	
Product Lifecycle Management	NOT STARTED	No model, page, or action found.	
Quality	DONE	Incoming/in-process/final checks with defect tracking.	

7.0 HRM

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Employees	PARTIAL	Rich database + bulk import; personal/professional document upload is not implemented (only ID-number text fields, no file attachments beyond avatar).	
Recruitment	PARTIAL	Job postings + applicant pipeline are real and internal-only; no job-portal syndication.	THIRD-PARTY API
Leaves	DONE	Holiday calendar, types, balances, approval workflow.	
Attendance	PARTIAL	Clock in/out and a lat/long-capable API endpoint exist, but there is no actual geofence distance check anywhere — the location is stored, not validated.	MOBILE APP
Performance	DONE	Reviews + goals, bulk import.	
Scheduling	DONE	Real month/week calendar grid, not a list view.	
Fleet	DONE	Vehicles, trip approval workflow that auto-generates linked project expenses, fuel logs with photo capture.	

8.0 Sites / Projects

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Projects (tasks / milestones / timesheets / files)	DONE	Full CRUD plus a genuinely custom-built Gantt chart (no third-party Gantt library).	
Templates	DONE	Create/apply project templates.	
Timesheets (with manager approval)	DONE	Approval workflow present.	
Tickets	PARTIAL	Full CRUD and a public portal lookup page exist, but the UI is a filterable list, not the kanban board the spec calls for (a real drag-and-drop kanban does exist for project tasks, just not tickets).	
Field + Form	PARTIAL	A capable drag-and-drop form builder exists, but lives under Organization rather than Projects, and there is no dedicated onsite-operations scheduling module.	

9.0 Organization

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Business Portal	PARTIAL	Seat count, storage allocation, and per-category storage stats are real; subscription billing has no payment gateway wired (status defaults to "Not configured").	THIRD-PARTY API
Departments	DONE	Interactive hierarchy chart + employee assignment.	
Branches	DONE	Interactive hierarchy chart, bulk import.	
Structure / dynamic RBAC	DONE	Department → Designation → Role → Permission fully modeled and seeded across ~12 modules; matches the tracking sheet's own note that "the functionality has been made dynamic."	
Credentials	PARTIAL	Upload works via the generic Document model, but no file-size cap is enforced and files are stored as base64 text in Postgres, not object storage.	INFRASTRUCTURE
Approvals	PARTIAL	Workflow definitions (steps, JSON config) are real; there's no request/instance model, so it isn't actually wired to approve or reject real quotations, invoices, or expenses yet.	

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Notices	DONE	Broadcast announcements with priority levels.	
Calendar (Meetings / Appointments / Reminders)	PARTIAL	CRUD and reminders are real; Google/Outlook OAuth connect flow genuinely captures tokens, but the sync function that would push events is never actually called.	THIRD-PARTY API
Notes	DONE	To-do / note types with due dates.	
Signatures	DONE	Genuine in-house canvas draw pad with request → sign/decline workflow — no third-party e-sign vendor involved.	
Contracts	DONE	Keyword search + project linkage.	INFRASTRUCTURE
Library	DONE	Same Document model as Credentials, category/tag search.	
Reports (Project / HR / Assets / Vendor)	PARTIAL	Report builder only currently supports leads/deals/invoices/contacts/activities — the Project, HR, Assets, and Vendor report sources named in the spec aren't wired in yet.	
Database (health / activity)	DONE	Real internal record-count and audit-log health page.	

10.0 Office / Workspace

The strongest module in the audit — every sub-feature is genuinely implemented, not stubbed.

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Documents (Word-style)	DONE	Rich-text editor with ribbon UI and PDF export.	
Spreadsheets (Excel-style)	DONE	Hand-built formula engine — SUM, AVERAGE, IF, cell references, circular-reference detection.	
Presentations (PowerPoint-style)	DONE	Real export via pptxgenjs, theme support.	
Email (webmail client)	DONE	Real IMAP sync and SMTP send, credentials encrypted at rest, validated live before saving.	THIRD-PARTY API
Messaging (channels / DMs)	DONE	Full chat CRUD with reactions and edit/delete; updates via 4–15s polling rather than websockets — functional, not instant.	

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Calls (audio & video)	DONE	Genuine peer-to-peer WebRTC with DB-relayed signaling — no third-party video SDK needed for the core case.	INFRASTRUCTURE

11.0 Website

Every piece below is genuinely built on the authoring side. The consistent gap is the same across all four: there is no public-facing route that actually serves a tenant's content to an outside visitor.

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Website builder	PARTIAL	Real drag-and-drop visual editor; no route found that publishes a page for live visitors.	
Blog	PARTIAL	Fully modeled with SEO fields, slugs, comments; no public blog route.	
Forum / FAQ	PARTIAL	Upvote, pin, best-answer all work internally; no public route.	
Live chat	PARTIAL	Agent-side console is real; no embeddable widget script for actual site visitors to open a chat.	

12.0 Report Generator / Civil Module

Flagged for planning attention. This is the least-started major module in the audit. What exists is a generic, type-agnostic report template shell — the domain-specific calculation and instrument-integration logic the module name implies has not been started.

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Geotechnical reports (CBR, SPT)	NOT STARTED	No calculation logic anywhere; no matches for CBR, California Bearing, or SPT in source.	HARDWARE
Survey reports (Road / Site / Canal)	NOT STARTED	Only the generic template shell exists, no survey-specific fields.	
Design reports (Road / Building / Canal)	NOT STARTED	Same generic shell, no design-calculation logic.	

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Estimation (BOQ)	PARTIAL	BOQ line items (quantity/rate/amount) are real and tied to Tenders; no Schedule-of-Rates master list to look rates up from.	

13.0 Settings

Sign Up/Login items are covered under §1.0 above and not repeated here.

FEATURE	STATUS	EVIDENCE	DEPENDENCY
Roles & RBAC (Access, User Assignments)	DONE	Full permission-matrix UI backed by real Role/Permission/UserRole tables.	
Mail Server setup & allocation	DONE	Per-tenant IMAP/SMTP configuration; employee mail access is self-service rather than admin-allocated, which is a minor process difference from spec.	THIRD-PARTY API
Industry onboarding templates	DONE	10 industries / 23 sub-categories seeded, matching the tracking sheet's Industry tab.	
Organization — company info & compliance	PARTIAL	PAN, GST, CIN, and address fields are present and editable; TAN and DIN fields named in the spec are missing from the schema.	

A Third-party services & hardware — pending

Every item below is blocked (fully or partly) on something outside the platform itself: a vendor account, a government API, or a physical device. Each is marked for review — confirm owner, budget, and timeline before committing a delivery date around it.

Payment gateway — production keys Razorpay + Stripe client code is fully wired (invoices, event ticketing, org subscriptions) but running on test/no credentials. Type: Third-party API · Blocks: Sales payments, Marketing event ticketing, Organization business-portal billing	for review
--	------------

SMS gateway (e.g. Twilio, MSG91)

for review

No SMS provider integrated anywhere; SMS marketing campaigns currently no-op.

Type: Third-party API · **Blocks:** Marketing – SMS campaigns

WhatsApp Business API — verified account

for review

Meta Cloud API client code is built and functional; needs a verified Meta Business account and long-lived access token to send in production.

Type: Third-party API · **Blocks:** Marketing – WhatsApp campaigns

Social platform API credentials

for review

Facebook Page token, LinkedIn OAuth app, and X/Twitter API keys needed — posting code is written, credentials are not configured.

Type: Third-party API · **Blocks:** Marketing – Social publishing

Job portal APIs (Naukri, LinkedIn Jobs, Indeed, WorkIndia)

for review

Sheet itself flags this "to be discussed." No syndication code exists yet — job postings stay internal until an integration is scoped.

Type: Third-party API · **Blocks:** HRM – Recruitment job posting reach

Zomato / Swiggy order integration

for review

No aggregator API code exists; needs partner-API access from each platform.

Type: Third-party API · **Blocks:** Sales – POS order integration

GSTN GSP — e-invoicing & e-way bill

for review

Required to move Accounts Receivable from "GST fields present" to actual IRN-compliant e-invoicing.

Type: Government portal · **Blocks:** Finance – AR e-invoicing, GST engine

ICEGATE, MCA21, EPFO/ESIC portal access

for review

Needed for the "Regulatory & Government Portal Integration" line item and statutory payroll filings.

Type: Government portal · **Blocks:** Finance – regulatory integration, Payroll statutory filing

Credit bureau data feed

for review

Needed for risk scoring in Credit Management & Collection.

Type: Third-party API · **Blocks:** Finance – Credit management

Google / Microsoft calendar sync completion

for review

OAuth consent and token storage already work — this is a scoping/priority question on whether to finish the sync call, not a new credential.

Type: Third-party API · **Blocks:** Organization – Calendar external sync

TURN relay server (e.g. coturn, or a hosted TURN provider)

for review

Calls currently rely on peer-to-peer WebRTC over public STUN only — will fail to connect across strict corporate firewalls/NAT without a TURN relay as fallback.

Type: Infrastructure · **Blocks:** Office – reliable Calls across all networks

Object / blob storage (S3, Azure Blob, or GCS)

for review

Contracts, credentials, signatures, and fuel-log photos are currently stored as base64 text directly in Postgres with no size cap — works today, will not scale.

Type: Infrastructure · **Blocks:** Organization – Credentials/Contracts, HRM – Fleet photos, at scale

Companion mobile app (iOS / Android or PWA)

for review

Does not exist yet. A set of backend endpoints (attendance, leaves, inventory scan, tickets, projects) already appear pre-built for such an app, but nothing is consuming them yet.

Type: Mobile app · **Blocks:** Geofenced attendance, live Map tracking, biometric login, field data capture

Biometric attendance devices

for review

Fingerprint/face capture hardware at each site, contingent on the mobile app or a dedicated terminal existing first.

Type: Hardware · **Blocks:** HRM – Attendance verification

Geotechnical/survey testing instruments — data export format

for review

CBR/SPT rigs and survey instruments need an agreed data format before their readings can feed into any report generator.

Type: Hardware · **Blocks:** Report Generator – Geotechnical & Survey reports

POS peripherals (receipt/KOT printers, barcode scanners, cash drawers)

for review

No POS hardware driver/integration layer exists yet.

Type: Hardware · **Blocks:** Sales – POS day-to-day operation

Self-service kiosk terminals

for review

Tablet/touchscreen units needed to deploy the (not-yet-built) Kiosk Software feature.

Type: Hardware · **Blocks:** Sales – Kiosk software

B Open items — feedback log

From the shared tracking sheet's feedback log — items with no "done" marked as of the sheet's last update. Worth a joint pass to confirm which are still live.

AREA	ITEM
Payroll	Reports payslip generation "is not functioning" — unresolved, corroborated as a real gap by this audit (see §3.0).
Payroll	Requests a knowledge-transfer walkthrough on how payroll works.
Vendor Bill	Requests a knowledge-transfer walkthrough on vendor bills.
Financial reports	Requests a knowledge-transfer walkthrough on financial reports.
Clients & Vendors	A field requested when adding a contact — description needs re-clarification before it can be scoped.
Warehouse	Clicking product quantity doesn't display product type.
Leaves	Leave type isn't available to select for newly added employees.
Notices	Only HR can post holiday announcements; employees cannot. Confirm if this is by design or a gap.
Organization — Notices	Choosing "department" as audience doesn't reveal a department picker; posted notices don't appear in the dashboard bell icon.

C Technical risk notes

Three patterns recur often enough across the audit above that they're worth calling out once here rather than repeating in every row:

- **Print-dialog PDFs.** Quotations generate their "PDF" via the browser's native print dialog rather than a proper generated file, even though server-side PDF generation is already used elsewhere in the platform for Office document exports. Worth standardizing.
- **Wired-but-unused integrations.** Google/Outlook Calendar OAuth and the campaign-sending pipeline are both cases where the hard part — authentication, API client code — is done, but the final call that actually does the work was never wired up. These are typically fast to close relative to their audit status of "Partial."
- **Base64-in-Postgres file storage.** Documents, contracts, signatures, and fuel-log photos are all stored as base64-encoded text directly in the database, with no size limit enforced anywhere. It works at current data volumes; it will become a real performance and cost problem as usage grows, and should move to object storage before that happens.

Biswajit

Trackepay · Engineering Delivery

14 July 2026